

Investigation of a Van der Pol Oscillator in contact with a Chaotic Thermostat

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1 Introduction

Morales (2018) demonstrated that supplementing a deterministic thermostat with a self-consistent fluctuating force converts it into a chaotic heat bath capable of thermalizing a free particle. An initially cold particle self-heats, its velocity distribution relaxes toward a Maxwellian, and the frequency spectrum displays the exponential dependence characteristic of chaotic dynamics. This result was extended to two spatial dimensions using the Martyna–Klein–Tuckerman (MKT) thermostat.

The present investigation asks what happens when the particle being thermalized is not passive but is itself a self-oscillating nonlinear system — specifically, a Van der Pol (VdP) oscillator, whose autonomous dynamics drive it toward a stable limit cycle. The MKT thermostat and the VdP nonlinearity thus impose two competing organizing principles on the same coordinate. The bath drives the system toward a Maxwellian velocity distribution and an exponential power spectrum, while the VdP oscillator drives it toward a fixed-amplitude limit cycle with a discrete harmonic spectrum. The purpose of this summary is to characterize, case by case, which tendency prevails across the accessible parameter space.

2 Model and Scaling

Coupling a Van der Pol oscillator to a MKT bath of coupling time t_0 gives the physical equation of motion

$$\ddot{x} + \omega^2 x = -\mu_{\text{phys}} (1 - k_{\text{phys}}^2 x^2) \dot{x} - \gamma(t) \dot{x} + F(t),$$

with the bath variables (γ, η) and phase angles (θ, φ) evolving as in the MKT construction. The fluctuating force takes the form of

$$F = F_0(\sin \theta + \sin \varphi).$$

Time and frequency is scaled to the simple harmonic oscillator's own natural frequency.

$$\begin{aligned} \tau &= \omega t, & \xi &= \frac{x}{x_s}, & u &= \frac{v}{v_s}, \\ \Gamma &= \frac{\gamma}{\omega}, & \beta &= \frac{\eta}{\omega}, & f_0 &= \frac{F_0}{F_s}. \end{aligned}$$

The scaled equations of motion actually implemented in the numerical integrator take the form

$$\frac{d\xi}{d\tau} = u, \quad (1)$$

$$\frac{du}{d\tau} = -\xi + f_0(\sin \theta + \sin \varphi) - \Gamma u + \mu(1 - k^2 \xi^2)u, \quad (2)$$

$$\frac{d\Gamma}{d\tau} = \omega_\tau^2(u^2 - 1) - \beta\Gamma, \quad (3)$$

$$\frac{d\beta}{d\tau} = \Gamma^2 - \omega_\tau^{-2}, \quad (4)$$

$$\frac{d\theta}{d\tau} = \Gamma, \quad (5)$$

$$\frac{d\varphi}{d\tau} = \beta. \quad (6)$$

After scaling, the system possesses three free parameters throughout. The parameter

$$\omega_\tau \equiv \omega t_0$$

is the ratio of the oscillator period to the bath coupling time. Small ω_τ corresponds to a fast, strongly coupled bath that acts many times per oscillator cycle, while large ω_τ corresponds to a slow, weakly coupled bath that allows many oscillator cycles to elapse before the bath responds. The parameter μ sets the strength of the Van der Pol nonlinear damping, and k sets the amplitude of the unperturbed VdP limit cycle. Six cases spanning this three-parameter space were integrated with a fourth-order Runge–Kutta scheme at $d\tau = 0.01$, each averaged over a 20×20 grid of initial phase angles (θ_0, φ_0) and several initial velocities, for ensembles of several thousand realizations.

3 The Van der Pol Oscillator as baseline

Before introducing the thermostat it is useful to characterize the isolated Van der Pol oscillator, since its spectral and statistical properties furnish the reference against which every subsequent case is compared. The time series of position and velocity, shown in Fig. 1, display the familiar jump from a small initial displacement to a fixed-amplitude, fixed-frequency limit cycle. The corresponding phase portrait in Fig. 2 shows the characteristic inward spiral onto a closed orbit. The frequency spectrum of the position of the nonlinear oscillator, displayed in Fig. 3 consists of a series of sharp, discrete harmonic peaks at the fundamental frequency and its odd multiples. The comparison to a pure sine wave shows that the fundamental frequency is close to the natural frequency of the corresponding simple harmonic oscillator with angular frequency of 1 rad/s.

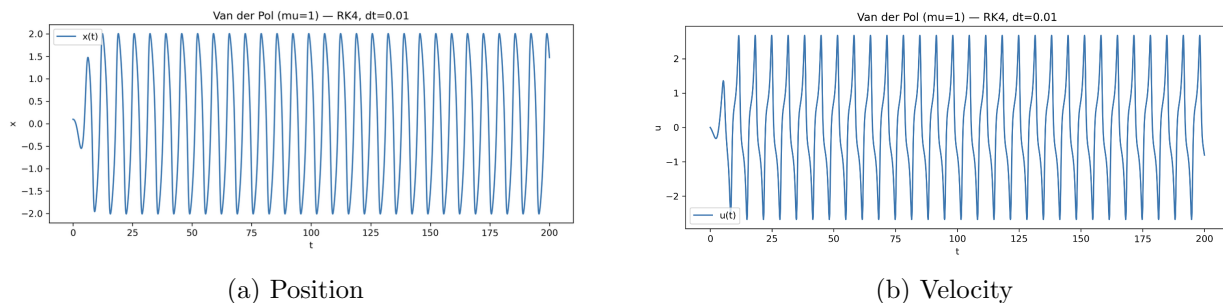


Figure 1: Position and velocity time series for VdP

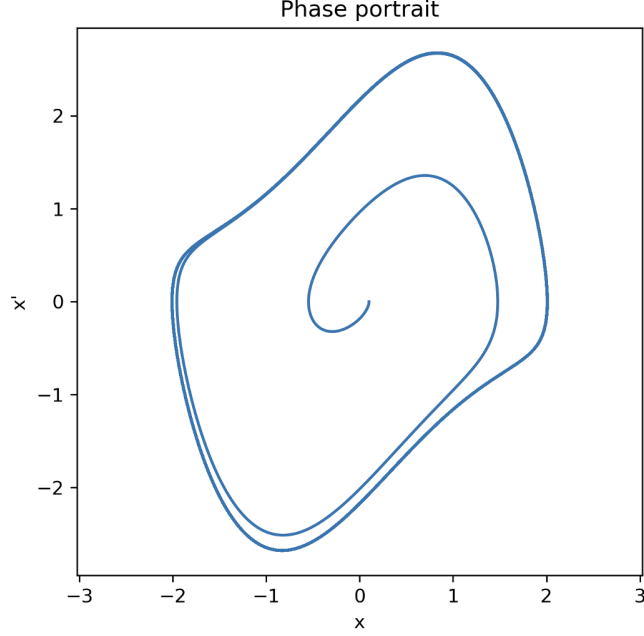


Figure 2: Phase Portrait for VdP

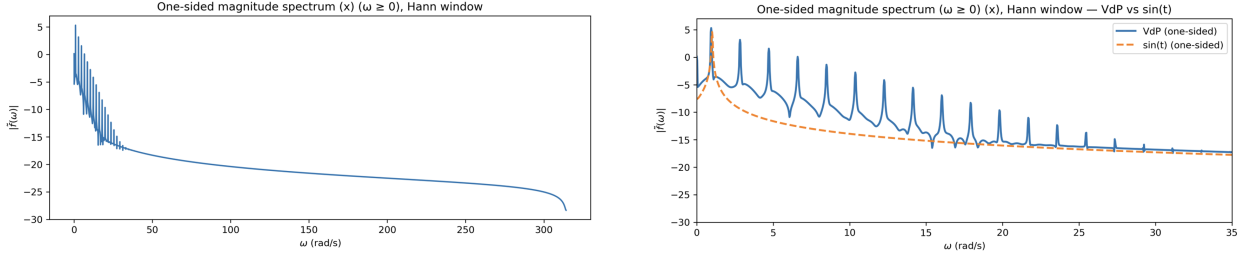


Figure 3: Log Linear FFT for position of VdP

4 Cases where $k = 0.5$

4.0.1 $\omega_\tau = 0.1$, $\mu = 0.1$, $k = 0.5$

At this combination of parameters the thermostat dominates completely (Fig. 4 and Fig. 5). The phase portrait is a filled, azimuthally isotropic disc. The velocity distribution is essentially an exact Maxwellian; the average position and velocity come out to be 1.01 and 1.12 respectively. This indicates that the system is almost thermalized. The frequency spectrum of ξ is a clean single-slope exponential. The Tarnopolski turning-point analysis places the coupled system's position curve close to the white-noise reference. At this combination of parameters the Van der Pol nonlinearity contributes essentially nothing; the system behaves as linear MKT bath. However, signatures of the Van der Pol oscillator and its nonlinearity are still visible in the frequency spectrum. A nonlinear Van der Pol oscillator typically produces a dominant fundamental frequency together with higher-frequency harmonics. This structure can be seen in Fig. 4, where small spectral peaks are superimposed on the otherwise decaying background. Although the semilog spectrum is dominated by an approximate straight line with a negative slope, indicating an exponential spectral decay, the initial peaks are representative of the underlying nonlinear oscillator. This feature is even clearer

in the linear-scale spectrum, where a pronounced low-frequency maximum is followed by a rapid exponential decay. The dominant peak, in both the position and velocity frequency spectrum, occurs at approximately $\omega \simeq 0.694 \text{ rad s}^{-1}$. Its position need not coincide exactly with the natural frequency of the uncoupled Van der Pol oscillator, since the observed dynamics result from the interaction between the nonlinear oscillator and the thermostat. Thus, even in this strongly thermalized regime, remnants of the intrinsic Van der Pol oscillator are visible.

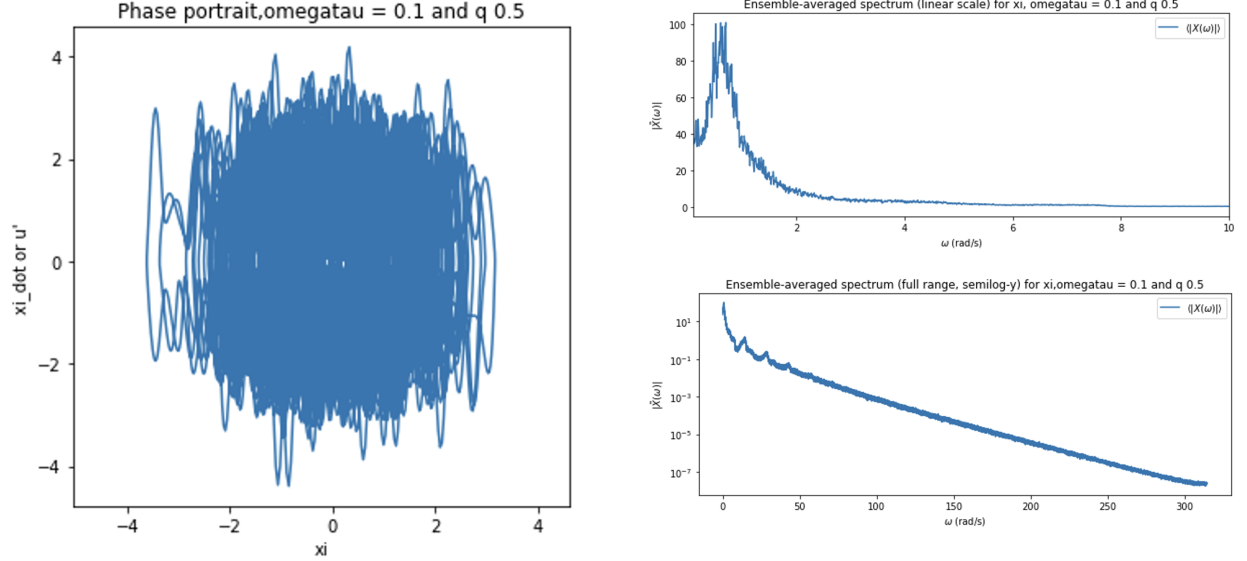


Figure 4: $\omega_\tau = 0.1$, $\mu = 0.1$, $k = 0.5$. (a) Phase portrait: a filled, isotropic disc. (b) Ensemble-averaged frequency spectrum of ξ .

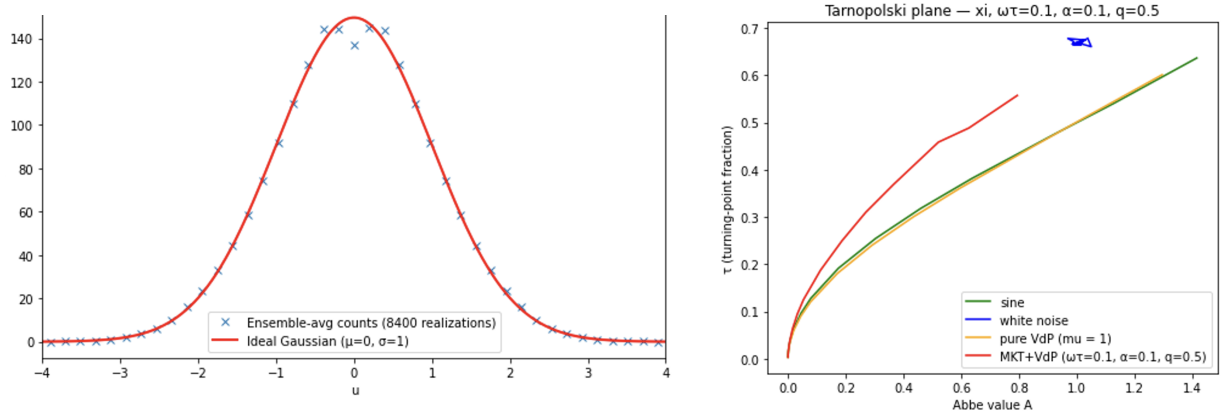


Figure 5: $\omega_\tau = 0.1$, $\mu = 0.1$, $k = 0.5$. (a) Ensemble-averaged velocity histogram (8400 realizations) against the ideal Maxwellian. (b) Tarnopolski turning-point plane, showing the position of the coupled system (red) above the sine/white-noise/pure-VdP references.

4.0.2 $\omega_\tau = 10$, $\mu = 1$, $k = 0.5$

Raising ω_τ to 10 while holding $\mu = 1$ and $k = 0.5$ reverses this outcome entirely (Fig. 6 and Fig. 7). The time series shows a regular, large-amplitude oscillation with no visible disturbance, and the phase portrait is a thick closed ring—a limit cycle broadened by residual chaotic forcing rather than

a filled disc. The velocity distribution is bimodal, reflecting the fact that the particle spends most of its time near the turning points of the limit cycle. On the linear scale, the spectrum is dominated by a sharp fundamental peak, accompanied by a sequence of smaller harmonics generated by the nonlinear oscillator. The dominant peak, in both the position and velocity frequency spectrum, occurs at approximately $\omega \simeq 0.962 \text{ rad s}^{-1}$. The running time averages confirm $\overline{v^2} \approx 7.5$ and $\overline{\xi^2} \approx 7.4$ —far above the thermal value of unity—while the damping coefficient Γ remains essentially undisturbed near zero throughout the run, indicating that the bath is simply too slow to act on the oscillator appreciably. The Tarnopolski curve for position lies essentially on top of the pure-VdP and sine reference curves. At large ω_τ the bath acts too slowly, relative to the oscillator period, to disturb the limit cycle appreciably, and the Van der Pol nonlinearity governs the dynamics almost entirely. Despite μ being ten times larger here than in the previous case, it is the change in ω_τ that has a significant impact.

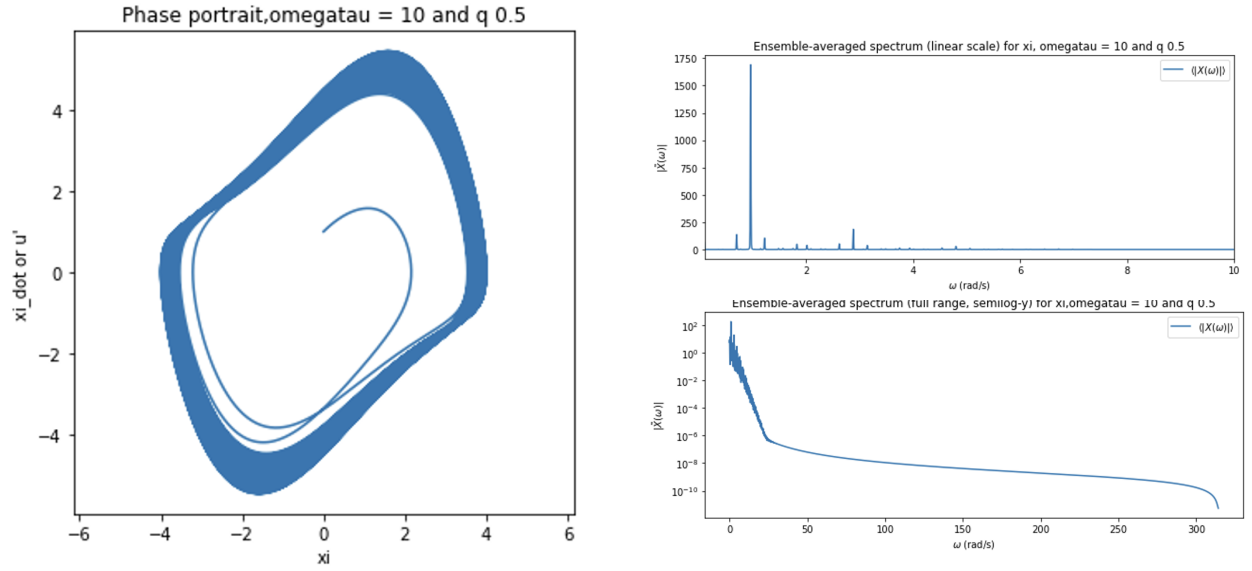


Figure 6: $\omega_\tau = 10$, $\mu = 1$, $k = 0.5$. (a) Phase portrait: a thick but closed limit-cycle ring. (b) Ensemble-averaged frequency spectrum of ξ .

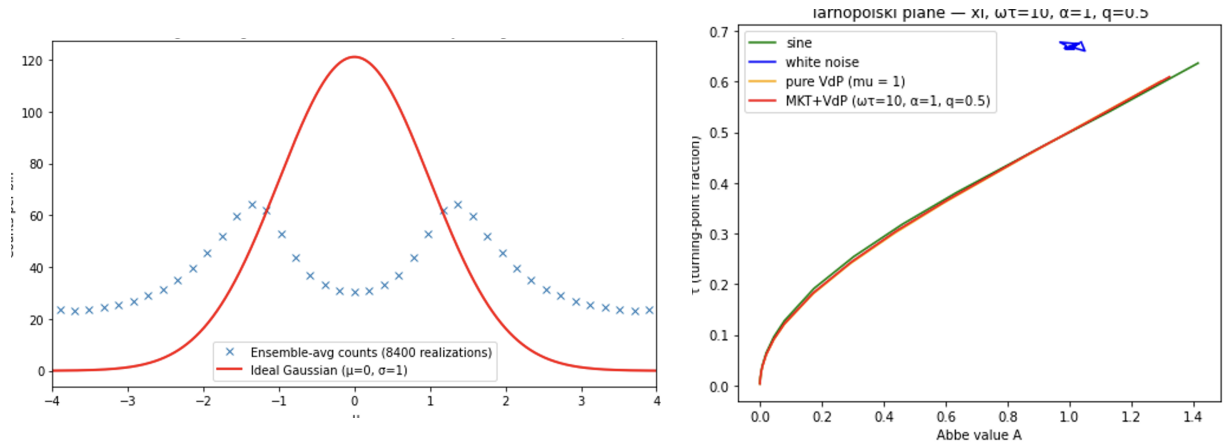


Figure 7: $\omega_\tau = 10$, $\mu = 1$, $k = 0.5$. (a) Velocity distribution. (b) Tarnopolski turning-point plane, showing the position of the coupled system relative to reference curves.

4.0.3 $\omega_\tau = 0.1$, $\mu = 15$, $k = 0.5$

The most informative of the three weakly nonlinear cases holds $\omega_\tau = 0.1$ fixed. However, μ is raised to 15 (Fig. 8 and Fig. 9). The time series is again essentially periodic, in sharp contrast to the noise-dominated trace obtained at the same ω_τ with $\mu = 0.1$, and the phase portrait shows a well-formed Van der Pol loop with fine-scale jitter superposed rather than either a filled disc or a perfectly clean ring. On the linear scale of the frequency spectrum, a dominant fundamental peak is visible together with several weaker secondary peaks, indicating that Van der Pol oscillations remain present. On the semilog- y scale, these low-frequency spectral features are superimposed on a broad background that decays steadily over higher frequencies. The spectrum, therefore, retains distinct oscillatory structure while also exhibiting substantial chaotic content introduced by the thermostat. The dominant peak, in both the position and velocity frequency spectrum, occurs at approximately $\omega \simeq 1.068 \text{ rad s}^{-1}$. The velocity distribution is noisy and only weakly bimodal, matching neither reference cleanly. The running time averages of $\langle \xi^2 \rangle$ and $\langle u^2 \rangle$ settle at values well above unity, consistent with the limit-cycle amplitude now dominating the energy budget rather than the bath's thermal scale. This case demonstrates directly that the nonlinearity parameter μ can override the influence of ω_τ . Even in a regime that strongly favors the thermostat, a sufficiently strong restoring nonlinearity retains Van der Pol like statistics.

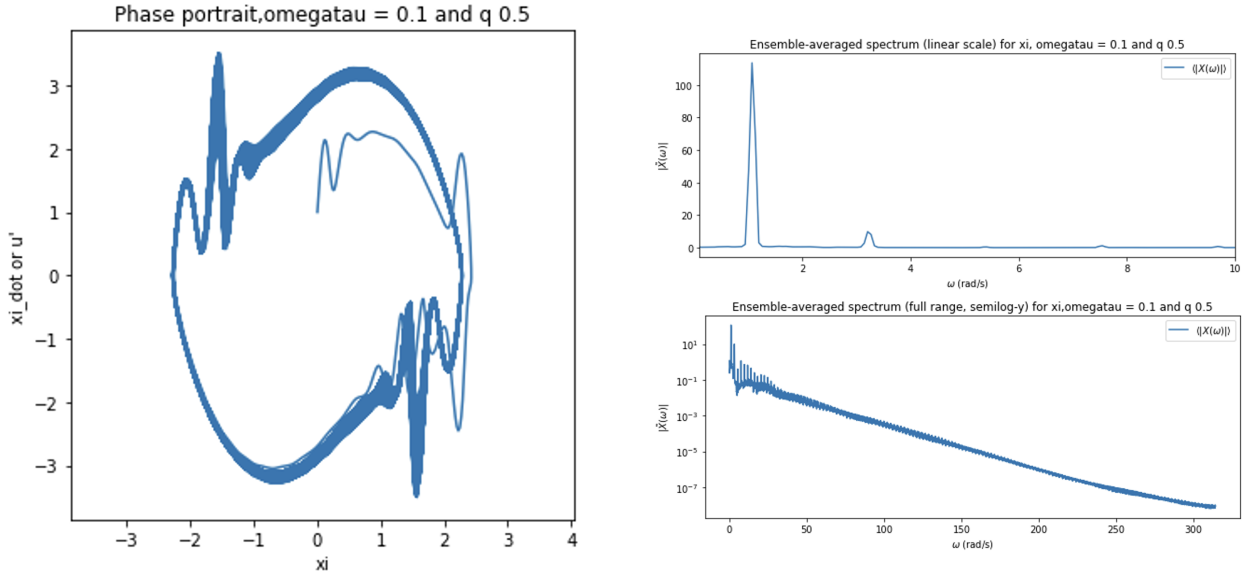


Figure 8: $\omega_\tau = 0.1$, $\mu = 15$, $k = 0.5$. (a) Phase portrait showing a Van der Pol-like loop with superposed jitter. (b) Ensemble-averaged frequency spectrum of ξ on linear and semilog- y scales. The linear spectrum displays pronounced discrete harmonic peaks, while the semilog representation shows these peaks superimposed on a broadband background.

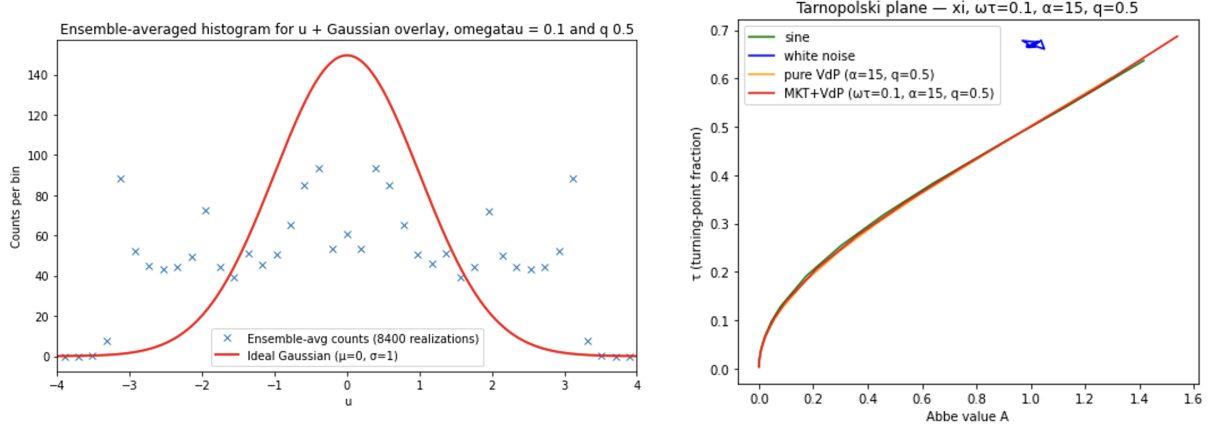


Figure 9: $\omega_\tau = 0.1$, $\mu = 15$, $k = 0.5$. (a) Ensemble-averaged velocity histogram, showing a noisy and weakly bimodal distribution. (b) Tarnopolski turning-point plane of ξ , showing the coupled system close to the pure-VdP reference.

5 Cases where ($k = 7$)

5.0.1 $\omega_\tau = 0.1$, $\mu = 1$, $k = 7$

Holding $\omega_\tau = 0.1$ and $\mu = 1$ and raising k to 7 produces a dense, self-intersecting phase portrait that resembles neither a filled disc nor a clean limit cycle (Fig. 10 and Fig. 11). The velocity distribution is sharply peaked and much narrower than the reference Maxwellian, consistent with the low final energy reported for this case ($\overline{v^2} = 0.305$). The zoomed time series makes this intermediate character particularly clear. The motion remains strongly oscillatory and retains a recognizable cycle; however, successive oscillations do not repeat periodically. Both, the amplitude and waveform vary gradually from one cycle to the next. The system is, therefore, ‘weakly chaotic’. The frequency spectrum supports this interpretation. On the linear scale, a dominant peak is visible at low-frequencies, indicating that a preferred oscillation is still present. However, the peak is followed by an exponential decay extending to higher frequencies. This differs from a clean limit cycle, which would produce sharply localized fundamental and harmonic peaks. The dominant peak, in both the position and velocity frequency spectrum, occurs at approximately $\omega \simeq 0.982 \text{ rad s}^{-1}$. Even the Tarnopolski curve for position tracks the Van der Pol and sine references closely at fine subsampling. However, it bows upward at coarse subsampling, suggesting a chaotic contribution that manifests only on long timescales.

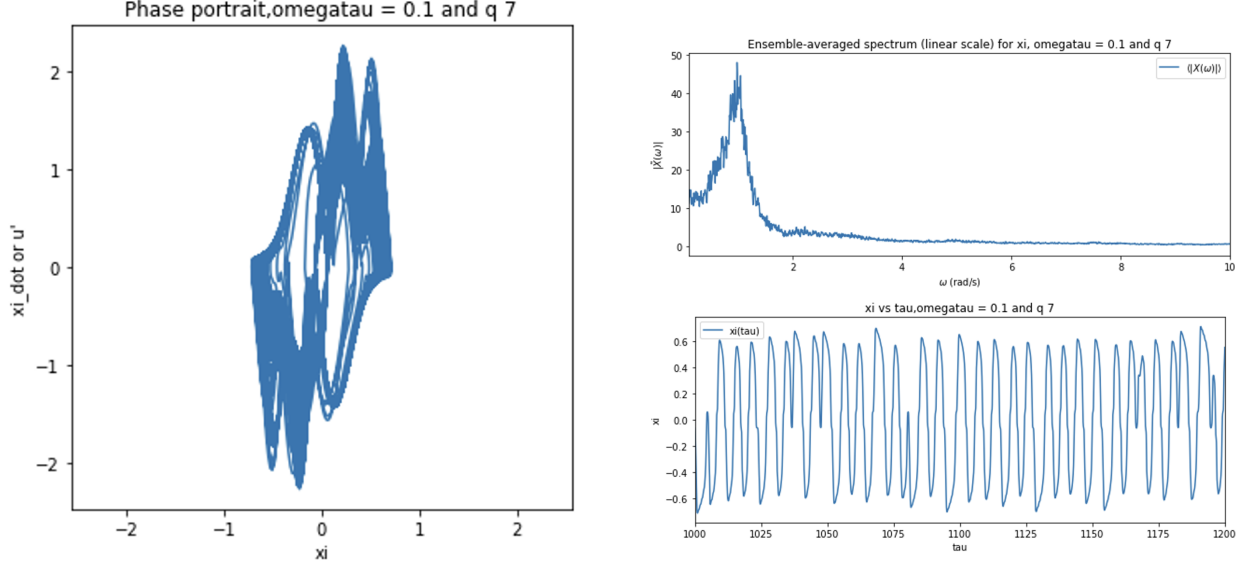


Figure 10: $\omega_\tau = 0.1$, $\mu = 1$, $k = 7$. (a) Phase portrait showing a dense, self-intersecting orbit. (b) Frequency spectrum of ξ , followed by a zoomed time series showing a recognizable oscillatory cycle with cycle-to-cycle variations in amplitude and waveform rather than exact periodicity.

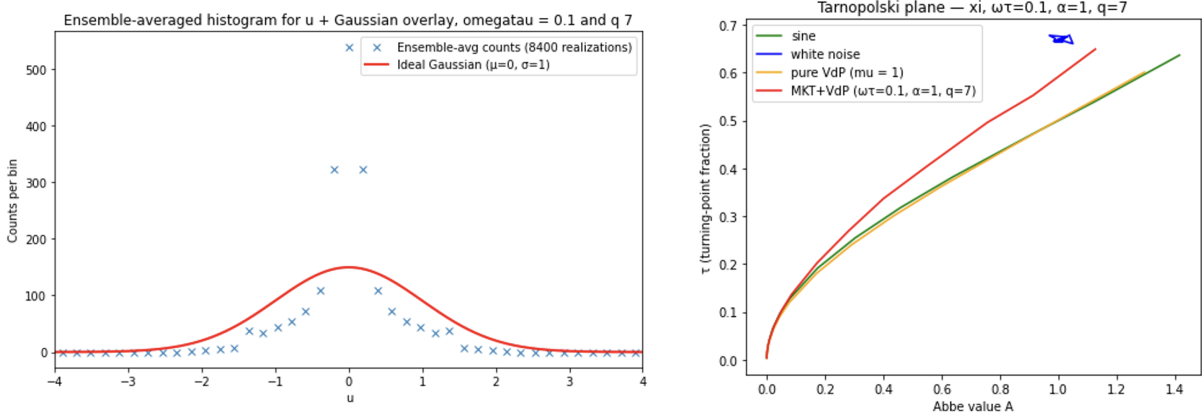


Figure 11: $\omega_\tau = 0.1$, $\mu = 1$, $k = 7$. (a) Ensemble-averaged velocity distribution compared with the ideal Maxwellian. (b) Tarnopolski turning-point plane, showing the coupled-system curve close to the Van der Pol and sine references at fine subsampling, with increasing separation at larger scales.

5.0.2 $\omega_\tau = 0.1$, $\mu = 7$, $k = 7$

Raising μ from 1 to 7 at the same $\omega_\tau = 0.1$ and $k = 7$ produces the most visually striking case in the entire set (Fig. 12 and Fig. 13). The phase portrait displays a near-perfect, thin, closed limit cycle—the cleanest and most deterministic orbit obtained at this value of ω_τ . The frequency spectrum, however, reveals that the dynamics are not entirely periodic. On the full-range semilog- y representation, the spectrum follows an approximately straight downward trend over a substantial frequency range, indicating an approximately exponential decay of spectral amplitude. Since this decay dominates the full-range view, the discrete oscillatory peaks are not immediately obvious on the semilog scale. The linear-scale spectrum over the low-frequency range resolves this structure. A strong fundamental peak is present together with a sequence of weaker secondary peaks, consistent

with harmonics generated by the nonlinear Van der Pol oscillation. In fact, the dominant peak, in both the position and velocity frequency spectrum, occurs at approximately $\omega \simeq 1.322 \text{ rad s}^{-1}$. The two representations therefore reveal complementary aspects of the same motion. The sharp fundamental and secondary spectral peaks are consistent with the nearly periodic limit cycle seen in the phase portrait, whereas the broad decaying background indicates that additional irregular fluctuations remain present. Thus, this case is best described as a strongly coherent nonlinear oscillation in an extremely weak chaotic environment. The Tarnopolski curve is also consistent with this interpretation. At small subsampling scales the coupled-system curve remains close to the Van der Pol and sinusoidal references, reflecting the dominance of the coherent periodic component. At larger subsampling scales it separates substantially from these references, suggesting that the non-periodic contribution becomes increasingly important over longer timescales.

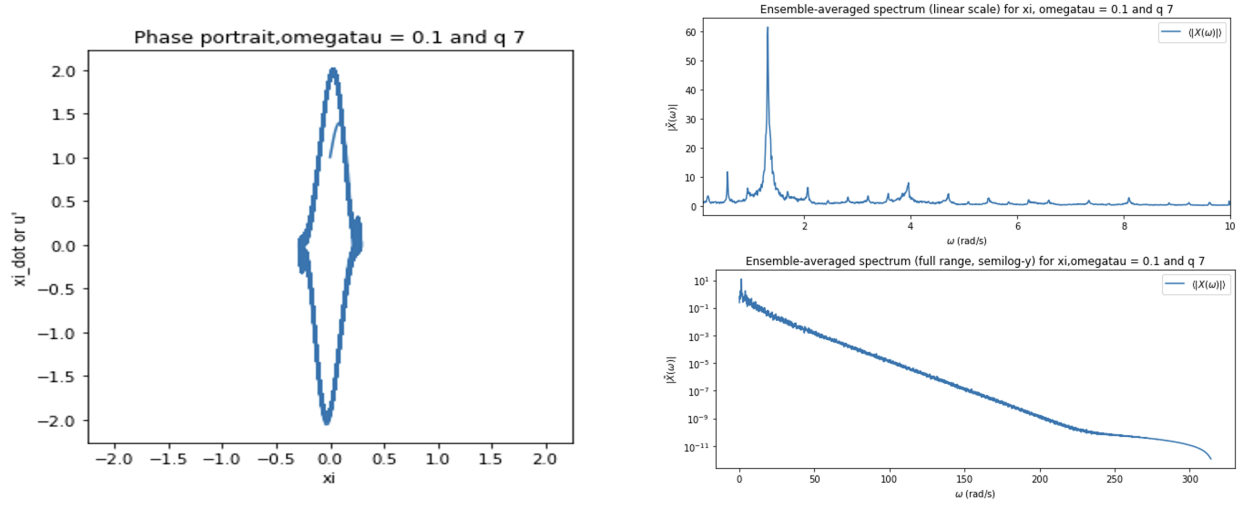


Figure 12: $\omega_\tau = 0.1$, $\mu = 7$, $k = 7$. (a) Phase portrait showing a thin, nearly closed limit-cycle orbit. (b) Ensemble-averaged frequency spectrum of ξ shown on linear and semilog- y scales.

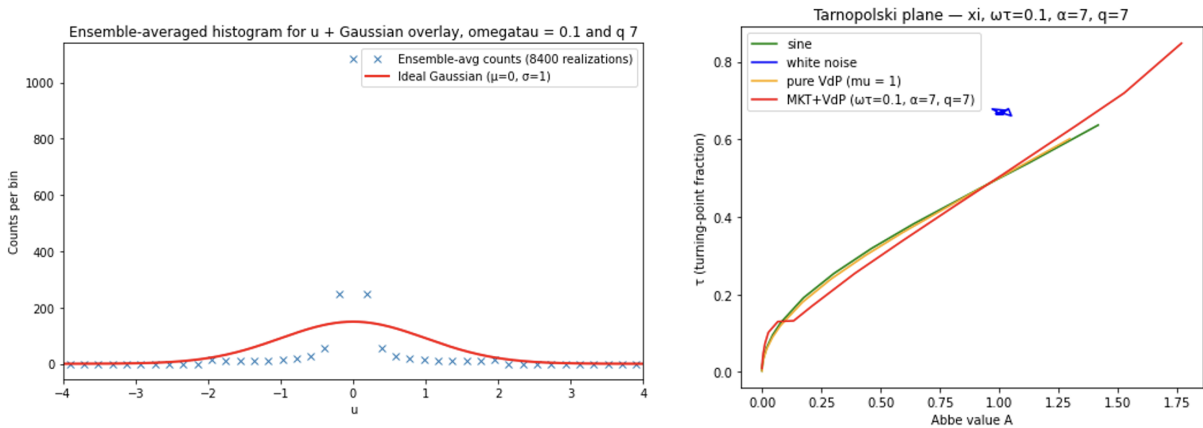


Figure 13: $\omega_\tau = 0.1$, $\mu = 7$, $k = 7$. (a) Ensemble-averaged velocity distribution compared with the ideal Maxwellian. (b) Tarnopolski turning-point plane of ξ : the coupled-system curve remains close to the Van der Pol and sinusoidal references at small subsampling scales, but separates from them at larger scales.

5.0.3 $\omega_\tau = 10$, $\mu = 7$, $k = 7$

Finally, at $\omega_\tau = 10$, $\mu = 7$, and $k = 7$, the system settles into a textbook relaxation oscillation (Fig. 14). The phase portrait shows flat plateaus near zero velocity punctuated by sharp spikes, tracing an essentially closed orbit of narrow width. The velocity distribution is a narrow, tall spike at zero, the most extreme departure from a Maxwellian obtained in any case, and the frequency spectrum is a clean discrete comb of harmonics with almost no chaotic component—the purest Van der Pol spectral signature obtained in this study. The dominant peak, in both the position and velocity frequency spectrum, occurs at approximately $\omega \simeq 0.100 \text{ rad s}^{-1}$.

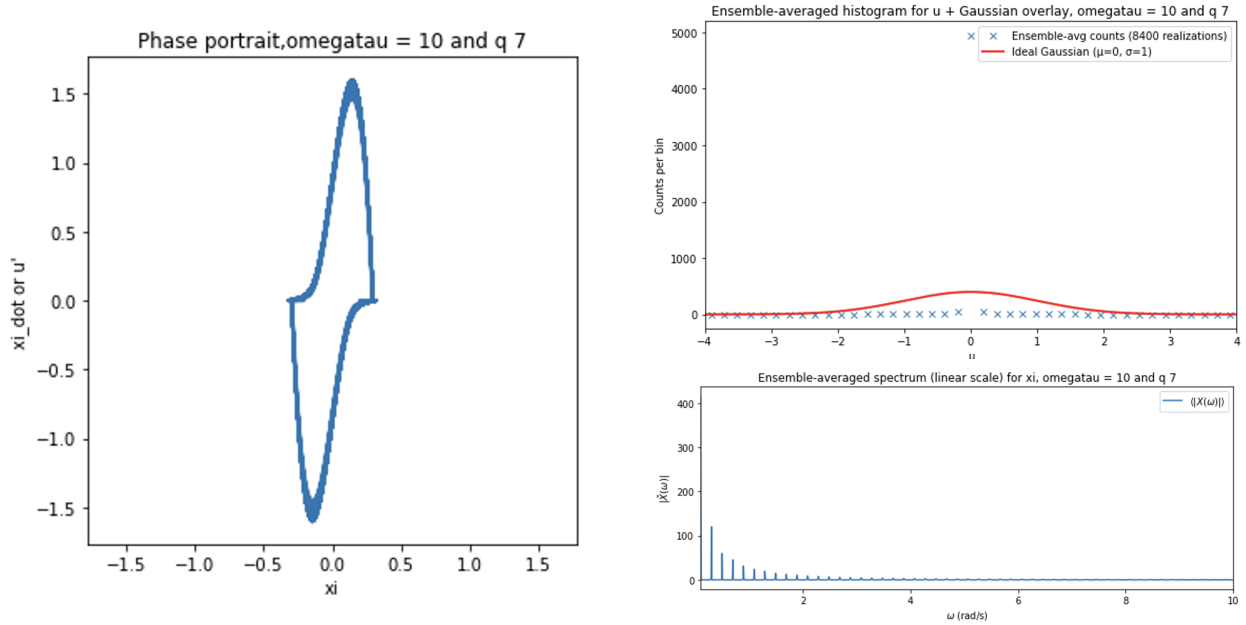


Figure 14: $\omega_\tau = 10$, $\mu = 7$, $k = 7$. (a) Phase portrait showing a thin, closed relaxation-oscillator loop. (b) Ensemble-averaged velocity distribution, strongly concentrated near $u = 0$, together with the corresponding linear-scale frequency spectrum showing a clean sequence of discrete spectral peaks.

6 Discussion

Considered together, the six cases indicate that no single parameter governs the outcome of the competition in isolation. The behavior at fixed $\omega_\tau = 0.1$ ranges continuously from unambiguously thermostat-dominated ($\mu = 0.1$) through mixed ($\mu = 1, 7$) to visually Van der Pol-dominated ($\mu = 15$, and again $\mu = 7$ at $k = 7$), as μ is increased. This suggests that the transition between thermostat-dominated and Van der Pol-dominated behavior is not governed by ω_τ alone, but by the relative strength of the two competing restoring mechanisms—the bath’s drive toward thermal equilibrium and the oscillator’s drive toward a fixed-amplitude limit cycle.